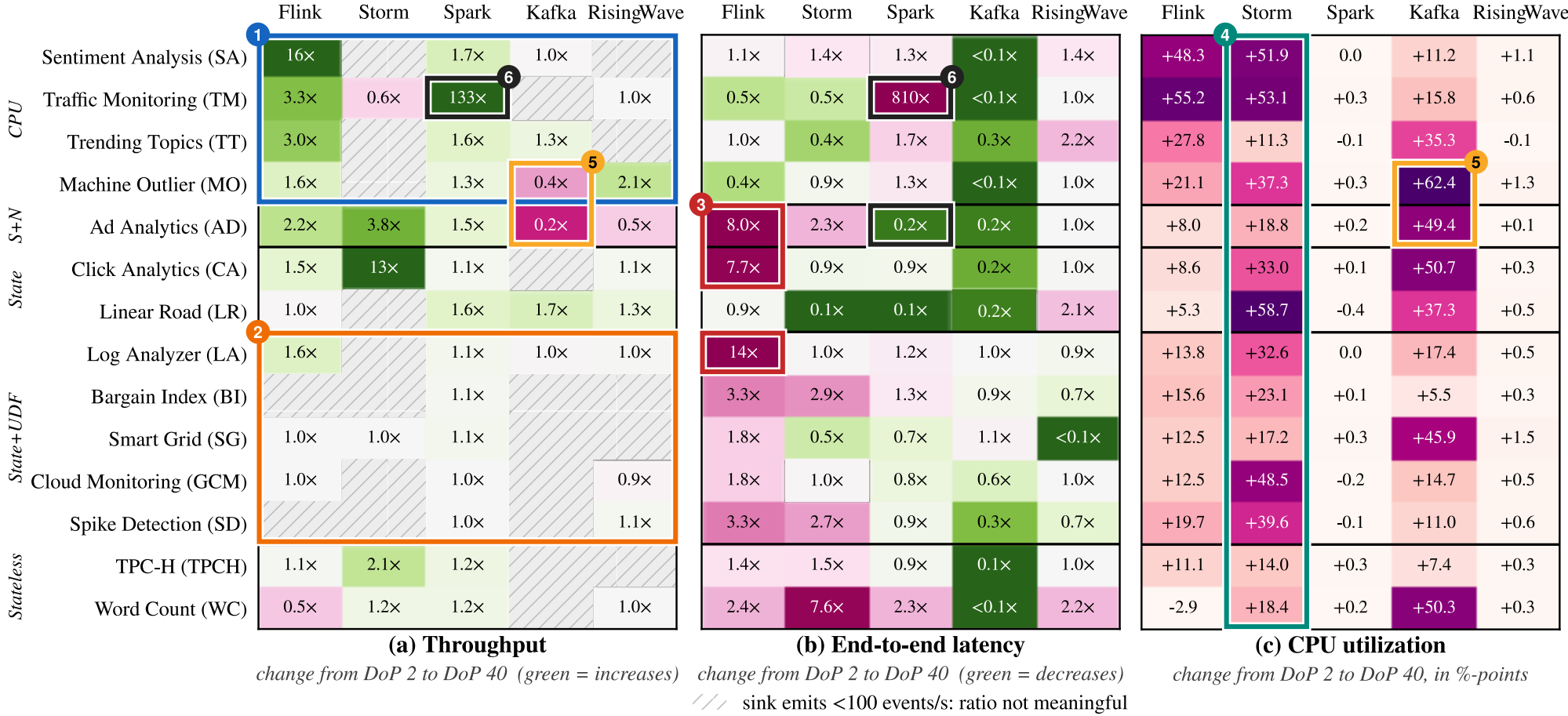

Flink Storm Spark Kafka RisingWave
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CPU
S+N
State
State+UDF
Stateless
Sentiment Analysis (SA)
Traffic Monitoring (TM)
Trending Topics (TT)
Machine Outlier (MO)
Ad Analytics (AD)
Click Analytics (CA)
Linear Road (LR)
Log Analyzer (LA)
Bargain Index (BI)
Smart Grid (SG)
Cloud Monitoring (GCM)
Spike Detection (SD)
TPC-H (TPCH)
Word Count (WC)
16× 1.7× 1.0×
3.3× 0.6× 133× 1.0×
3.0× 1.6× 1.3×
1.6× 1.3× 0.4× 2.1×
2.2× 3.8× 1.5× 0.2× 0.5×
1.5× 13× 1.1× 1.1×
1.0× 1.6× 1.7× 1.3×
1.6× 1.1× 1.0× 1.0×
1.1×
1.0× 1.0× 1.1×
1.0× 1.0× 0.9×
1.1×
1.1× 2.1× 1.2×
0.5× 1.2× 1.2× 1.0×
1.1× 1.4× 1.3× <0.1× 1.4×
0.5× 0.5× 810× <0.1× 1.0×
1.0× 0.4× 1.7× 0.3× 2.2×
0.4× 0.9× 1.3× <0.1× 1.0×
8.0× 2.3× 0.2× 0.2× 1.0×
7.7× 0.9× 0.9× 0.2× 1.0×
0.9× 0.1× 0.1× 0.2× 2.1×
14× 1.0× 1.2× 1.0× 0.9×
3.3× 2.9× 1.3× 0.9× 0.7×
1.8× 0.5× 0.7× 1.1× <0.1×
1.8× 1.0× 0.8× 0.6× 1.0×
3.3× 2.7× 0.9× 0.3× 0.7×
1.4× 1.5× 0.9× 0.1× 1.0×
2.4× 7.6× 2.3× <0.1× 2.2×
+48.3 +51.9 0.0 +11.2 +1.1
+55.2 +53.1 +0.3 +15.8 +0.6
+27.8 +11.3 -0.1 +35.3 -0.1
+21.1 +37.3 +0.3 +62.4 +1.3
+8.0 +18.8 +0.2 +49.4 +0.1
+8.6 +33.0 +0.1 +50.7 +0.3
+5.3 +58.7 -0.4 +37.3 +0.5
+13.8 +32.6 0.0 +17.4 +0.5
+15.6 +23.1 +0.1 +5.5 +0.3
+12.5 +17.2 +0.3 +45.9 +1.5
+12.5 +48.5 -0.2 +14.7 +0.5
+19.7 +39.6 -0.1 +11.0 +0.6
+11.1 +14.0 +0.3 +7.4 +0.3
-2.9 +18.4 +0.2 +50.3 +0.3
1
2
3
4
5
6
(a) Throughput
change from DoP 2 to DoP 40 (green = increases)
(b) End-to-end latency
change from DoP 2 to DoP 40 (green = decreases)
(c) CPU utilization
change from DoP 2 to DoP 40, in %-points
sink emits <100 events/s: ratio not meaningful